

Key: Who Moved, and Who Never Left

84-355 Democracy's Data | Migration | Instructor copy — do not distribute

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All code is given to students. Grade the interpretation.

Structure. The brief (`migration-brief.Rmd`) is now a chapter rather than a parallel copy of the lab: prose organised by argument, under idea headings, with seven numbered figures and no numbered steps. The lab (`migration.Rmd`) still runs Steps 1–11 with the code exposed. **Students should read the brief first.** Step numbers below refer to the lab; the chapter is cited by figure number or section title.

This is the course's margin-of-error session. Every other lab shows students a number somebody published. This one shows them the interval the same publisher printed beside it, and what happens to a famous-looking map when you honour it.

Every number below is computed from the built CSVs at knit time. Nothing in this document is typed in.

The session in one sentence

State-to-state migration is a survey estimate, not a count, and when you apply the margins of error the Census Bureau publishes beside it, 35% of the flows on the map turn out to be indistinguishable from nobody moving at all — and they are exactly the thin, surprising, interesting-looking ones.

The findings

```
c(state_pairs      = nrow(fl),
  distinguishable   = sum(fl$sig),
  pct_surviving     = round(100 * sum(fl$sig) / nrow(fl), 1),
  pct_of_movers_lost = round(100 * sum(fl$est[!fl$sig], na.rm = TRUE) /
                             sum(fl$est, na.rm = TRUE), 2),
  net_not_sig_states = sum(!S$net_sig),
  focus_net         = F1$net,
  focus_inflows_ok  = F1$sig_in_n,
  contrast_inflows_ok = C1$sig_in_n,
  county_pct_surviving = round(100 * mn("county_all_sig") /
                               mn("county_all_pairs"), 1))
```

```

#>      state_pairs      distinguishable      pct_surviving
#>      2652.00      1714.00      64.60
#>  pct_of_movers_lost  net_not_sig_states      focus_net
#>      2.21      22.00      -254332.00
#>      focus_inflows_ok  contrast_inflows_ok  county_pct_surviving
#>      51.00      14.00      27.40

```

- **938 of 2,652 state pairs (35%) cannot be told apart from zero** — and they hold only 2.21% of the movers. The uncertainty sits where the pattern is, not where the volume is.
- **California is down 254,332 net to other states**, which is 9.4 times its own margin of error and therefore real. **22 of the 51 jurisdictions have a net figure that is not.**
- **California can use 51 of its 51 inflows; Delaware can use 14.** Same survey, same year. Log population and usable inflows correlate at $r = 0.91$ — a flow map of a small state is mostly a picture of the sample size.
- **At county level only 27.4% of pairs survive** (338 of 1,234 feeding Los Angeles County), and that is after pooling five years of sample rather than one.
- **Born-in-state runs from 28.5% (Nevada) to 77.0% (Louisiana)**, nationally 56.5%.
- **Americans move at 46% of the 1948 rate**; interstate moving is at 42% of it.

The three things to get said

1. A margin of error is not decoration. Step 5 is the whole session. An estimate smaller than its own margin of error is a number the survey cannot distinguish from nobody. The published example to put on the board is **Minnesota to Utah: $2,617 \pm 3,046$** — the largest flow in the country whose own publisher says it is consistent with zero. **Make somebody say what a newspaper should print instead.** The honest sentence is “somewhere between 0 and 5,663, and we cannot rule out none.”

2. The uncertainty is not spread evenly, and that is what makes it dangerous. The failed pairs hold 2.21% of the movers: median surviving flow 1,779 people, median failed flow 129. So the map is trustworthy about volume and unreliable about *connections* — and the long thin exotic arc that makes a flow map worth looking at is precisely the arc that is not there. **The more distinctive a migration claim sounds, the more likely it is an artefact of sample size.**

3. Step 6 is the political consequence. California gets 51 usable inflows and Delaware gets 14, and nothing about that is a fact about the states. It is a fact about how many of their movers landed in the sample. **Ask what it means that the states whose migration we can describe confidently are the large ones**, when the states that gain and lose House seats at the margin are often not.

The line for the board: a survey that cannot see a flow has not measured zero, and a map that draws it anyway is drawing its own sample size.

Walking the steps

Step	What it does	Min	Watch for
1 What a statistic is	One question: where did you live a year ago	5	Most moving is local. Interstate is the small slice everyone argues about.
2 One state	California in, out, net, abroad	5	The $\pm$ column is already on screen. Nobody reads it yet. That is the setup.
3 The arc map	Population-weighted centres, both directions	6	The heaviest arc runs both ways. Net is a difference of two large numbers.
4 Every state	Net per 1,000, ranked	5	Grey bars are already visible here. Do not explain them yet.
5 None are counts	Apply the published margins	12	The pivot. Take the prediction first. Protect this step.
6 Whose map survives	51 vs 14 usable inflows	8	Size, not connectedness. The scatter makes it mechanical.
7 Born in state	Stocks, not flows	6	28.5% to 77.0%. A century of migration, not this year's.
8 From abroad	A residence category, not a legal one	5	Say this precisely or not at all. Emigration is absent, not small.
9 One county	27.4% survive	5	Finer grain, worse estimates, <i>and</i> a different instrument.
10 The back-series	CPS, 1948–2020	5	Different survey. Do not put it on an ACS axis.
11 What you can say	Including what you cannot	4	“Why” is not in the file. Neither is anyone who left.

Where the chapter sits against that table. The rows are the lab’s steps and the minutes are for running the lab, so they stand as they are. The one place the two documents do not line up is rows 2 and 3: the chapter treats one state and its arc map as a single section, *The state with the largest imbalance*, ending in Figure 1, so a class working from the chapter will not find a seam between them. Run the pair as one eleven-minute block and take the break wherever the room’s attention gives out; the lab still has a **# Step 3** heading if you want the old cut. The rest map one to one: row 1 to *Nobody votes on this* and *Most moving is local*, row 4 to Figure 2, row 5 to *An estimate smaller than its own margin of error* and Figure 3, row 6 to *Whose map can be drawn at all* and Figures 4–5, row 7 to Figure 6, row 8 to *A residence category, not a legal one*, row 9 to *Finer grain, worse estimate*, row 10 to Figure 7, and row 11 to *What a survey of movers can testify to*.

Step 5 deserves the prediction, and it is the best prediction in the course. Ask the room: *there are 2,652 ordered pairs of states — how many describe a flow the survey can tell apart from zero?* Almost everyone guesses something in the nineties. It is 65%. Then ask the follow-up before revealing it: *and how many of the actual movers are in the pairs that failed?* Having just been told a third of the map is unusable, the room expects a large number. It is 2.21%. **Both surprises are needed** — the first says the map is mostly noise, the second says the noise is weightless, and the interesting claim lives in the gap between them.

If you are short on time, protect Steps 5, 6 and 9. Everything else is context for those three.

Handling the room

- **Nobody writes code.** Say it at the start; the lab is dense with output and a few students read that as a coding session.
 - **The Big Sort is the hook.** Most students have met the claim that Americans are sorting themselves politically. The chapter's *The argument this data cannot settle* is where to spend any spare minutes: Mummolo and Nall, Martin and Webster, Chen and Rodden.
 - **Do not let the room settle the sorting argument.** This data has no political variable at all. The honest contribution is Step 10's constraint: whatever sorting is happening, it is happening while the mechanism runs at 42% of its 1948 speed.
 - **Step 8 is where a class can go wrong fast.** "From abroad" is a statement about geography. It counts returning citizens, military families, students and people arriving from Puerto Rico and the Island Areas alongside immigrants, and the file separates none of them. Say so before anyone infers otherwise.
-

Option A — Audit a state's map

Verified output

```
rbind(head(S[order(-S$sig_in_n), c("state", "pop1", "sig_in_n", "sig_out_n")], 3),
      head(S[order( S$sig_in_n), c("state", "pop1", "sig_in_n", "sig_out_n")], 3))
```

```
#>      state      pop1 sig_in_n sig_out_n
#> 5   California 39039388      51      50
#> 10    Florida 23163294      50      49
#> 45     Texas 30944767      50      51
#> 8     Delaware 1043560      14      13
#> 35 North Dakota  787926      15      16
#> 41 Rhode Island 1103757      15      16
```

```
c(median_usable_inflows = median(S$sig_in_n),
   states_below_half    = sum(S$sig_in_n < NOR / 2),
   cor_logpop_usable    = round(cor(log(S$pop1), S$sig_in_n), 2))
```

```
#> median_usable_inflows      states_below_half      cor_logpop_usable
#>                35.00                11.00                0.91
```

3/4: Picks a state, reports its `sig_in_n` and `sig_out_n`, and says the map would be partly unusable.

4/4: Compares a large state to a small one and states the mechanism — the survey samples households, so a state with fewer residents contributes fewer movers to every cell, and the margin of error is large relative to

a small estimate. Says explicitly what the map would be honest about (total volume, the handful of heavy neighbouring flows) and what it would fake (every thin long-distance arc).

4/4, best: Notices that `sig_in_n` and `sig_out_n` are close for most states and asks why, or works out that **no state's flow map is complete**: even the median jurisdiction loses 16 of its 51 inflows, and 11 of 51 lose more than half. A student who converts this into a rule ("do not draw an arc map below population X") has understood the step better than the lab states it.

Option B — The league table nobody checks

Verified output

```
z <- S[!S$net_sig, c("state", "net", "net_moe", "net_per1k")]
z <- z[order(-abs(z$net)), ]
z$net_moe <- round(z$net_moe)
head(z, 8)
```

```
#>           state    net net_moe net_per1k
#> 39 Pennsylvania -17108  18778 -1.3207590
#> 25 Mississippi  10133   10781  3.4783677
#> 51 Wisconsin   9274   11756  1.5710914
#> 46 Utah        9266   12009  2.6789700
#> 26 Missouri    8061   13715  1.3040571
#> 32 New Mexico  -6844    8547 -3.2447814
#> 24 Minnesota   -6746   10721 -1.1774219
#> 23 Michigan    6121   14624  0.6093877
```

```
nrow(z)
```

```
#> [1] 22
```

3/4: Lists the states and reports that their net migration cannot be distinguished from zero.

4/4: Writes the corrected sentence. Something like: *Pennsylvania's population change from interstate migration in 2024 was -17,108, with a margin of error of $\pm 18,778$ — the survey cannot tell whether the state gained or lost.* The move that earns the 4 is refusing the direction, not just adding an interval.

4/4, best: Notices that these are mostly **mid-sized states** — not the smallest — and works out why: net is a difference of two large numbers, so its margin of error is the root of the sum of two squared margins and can easily exceed the difference itself. **A state can have perfectly well-measured inflows and outflows and a completely unmeasured net.** That is the sharpest available observation on the whole lab, and it explains why “winners and losers” stories are worse than the data they are built on.

Option C — Born here versus arrived

Verified output

```
round(cor(S$born_state_pct, S$net_per1k), 2)
```

```
#> [1] -0.25
```

```
r <- resid(lm(net_per1k ~ born_state_pct, data = S))
rbind(head(S[order(-r), c("state", "born_state_pct", "net_per1k")], 3),
      head(S[order( r), c("state", "born_state_pct", "net_per1k")], 3))
```

```
#>           state born_state_pct net_per1k
#> 47      Vermont      47.18941  14.182140
#> 29      Nevada      28.46887  12.759006
#> 42 South Carolina      50.68087  10.087620
#> 2        Alaska      43.01065 -14.525953
#> 31    New Jersey      50.45446  -6.798314
#> 43  South Dakota      61.86019  -7.979273
```

3/4: Reports the weak correlation and names a couple of states that do not fit.

4/4: Explains what the two variables are measuring on different clocks. Born-in-state share is a **stock** — the residue of every migration a state has ever had, plus its birth rate and its retention of the people born there. Net migration per 1,000 is a **flow**, one year wide. There is no reason for them to agree, and the weak correlation is the expected result rather than a puzzle.

4/4, best: Picks the outliers apart individually and finds that they are different stories: a state can be low on born-in-state and *losing* people (decades of churn that has stopped), or high on born-in-state and *gaining* (recent growth that has not yet worked through the stock). Best of all is the student who points out that the outlier list is unstable because 22 of the net figures cannot be told from zero anyway — **43% of the residuals in this scatter are noise, and the lab's own Step 5 says so.**

Option D — Their own question

No verified output. Grade the question as much as the answer.

3/4: A sensible question answered correctly from the files.

4/4: A question whose answer required checking a margin of error, or one that compares two things the lab kept separate and says why the comparison is limited.

4/4, best: A question the data **cannot** answer, correctly identified as such, with a statement of what source would be needed. Emigration and motive are the two obvious ones and both are named in the lab; credit anyone who finds a third.

Grading

Four-point scale from the syllabus.

- **Never penalise code.** It is given.
- **The move that earns a 4** is treating the margin of error as part of the finding rather than a caveat appended to it.
- **A 2** is “the data has margins of error so we can’t really know anything” — true-sounding, useless, and the opposite of what Step 5 shows. The point is not that nothing is knowable; it is that 97.8% of the movers are in flows we *can* measure, and the failures are concentrated somewhere specific.
- **Also a 2:** reporting any of these figures as a count of people.
- **Reward anyone who asks about the people who left the country.** Emigration is absent from this dataset, not small in it, and noticing the absence of a variable is harder than reading one that is present.
- **Reward anyone who separates “from abroad” from immigration** without being prompted.

Anticipated questions

- “*Why doesn’t the Census just count everybody?*” — It does, every ten years, and that count is what apportions the House. This is the survey that fills the nine years in between; annual migration is not something the decennial census measures at all.
- “*If the margin is that big, why publish the number?*” — Because the Bureau publishes the margin with it. The failure is downstream: newspapers, state development offices and think tanks republish the estimate and drop the interval. Nothing in this lab is a criticism of the Census Bureau.
- “*Couldn’t they just pool more years?*” — That is exactly what the county-to-county table does (2016-2020, five years), and even then only 27.4% of pairs survive. Pooling also costs you the ability to say *when*.
- “*Does this mean the flow maps I see online are wrong?*” — Not wrong about the heavy flows. Wrong to draw the thin ones with the same confidence, and usually silent about which is which.
- “*Why is Puerto Rico in the flow table but not the rankings?*” — It is an origin and destination for migration, so it belongs on the map, but its place-of-birth categories are constructed differently and it is not a state. The build script flags it rather than silently ranking it.
- “*Which state is really winning?*” — Only the ones whose net clears its own margin. 29 of 51 do; the top two are Texas and Florida. For the rest, the answer is that the survey does not know.

Connections

- **census and apportionment** — where migration ends up: seats. This lab is the nine years between the counts, and the reminder that the annual numbers everyone argues about are not the ones that move seats.
- **poll-simulation and gss-confidence** — the other places the course handles sampling error. Run one of them before this if you can; students who have already met a confidence interval get Step 5 much faster.
- **levels-of-aggregation** — the same lesson from the other direction: state totals are stable and county cells fall apart. Step 9 is that lab’s point with a published margin attached.
- **regional-shift** — seventy years of population moving between regions, measured by the census rather than a survey. Useful contrast: there, the number is a count and the hard part is what it means.
- **bisg-check** — the other lab where an estimate is routinely presented without the uncertainty its own authors published.